Algorithmic Objectivity for Congressional Redistricting: A National-Scale Demonstration

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Abstract

Congressional redistricting in the United States suffers a legitimacy crisis: partisan mapmakers manipulate district boundaries to predetermine election outcomes, eroding public confidence in representative democracy. We demonstrate that recursive graph bisection—a computational method used for supercomputer workload distribution— provides an algorithmic alternative that cannot be instructed to gerrymander because it cannot access partisan data. Applied to all 50 states across three census decades (2000/2010/2020), this method produces 435 congressional districts with near-perfect population equality, geographic contiguity, and geometric compactness. Algorithmic plans create 69 more majority-minority districts than enacted maps, exceeding Voting Rights Act requirements without explicit racial targeting. We identify a 42% demographic threshold determining where proportional minority representation becomes geographically feasible. These findings establish that mathematical governance—successful for congressional apportionment since 1941—can extend to redistricting, offering a path toward restoring electoral legitimacy through structural objectivity rather than institutional reform.

1 The Redistricting Crisis

Every ten years, American democracy faces a constitutional requirement with no constitutional solution: redrawing 435 congressional district boundaries to reflect population changes from the decennial census. This process, affecting 330 million Americans and determining electoral outcomes for a decade, suffers from a fundamental legitimacy crisis. Partisan legislatures manipulate district boundaries to predetermine election results—a practice known as gerrymandering—eroding public confidence in representative government [1].

The problem is ancient but newly precise. While gerrymandering dates to 1812, modern computing power enables manipulation with surgical accuracy [2]. Reformers have proposed independent redistricting commissions to remove partisan incentives, yet these still rely on human judgment about what constitutes "fair" districts. Even well-intentioned commissioners must balance competing values—compactness, community preservation, competitiveness—with no objective formula to guide their choices [3].

The legal landscape offers no remedy. In *Rucho v. Common Cause* (2019), the Supreme Court ruled that partisan gerrymandering claims present "political questions beyond the reach of federal courts," effectively removing federal oversight [4]. Chief Justice Roberts acknowledged the practice "incompatible with democratic principles" but declared courts institutionally incapable of policing it. This ruling created a governance vacuum: states may gerrymander with impunity, and no mathematical standard exists to distinguish legitimate from illegitimate maps.

This program is scoped to congressional and state-legislative redistricting. Local redistricting for city councils, school boards, and county commissions is outside the present research boundary.

We demonstrate that recursive graph bisection—a computational method developed for distributing workloads across supercomputer processors—provides structural objectivity for redistricting. Applied to all 50 states across three census decades (2000, 2010, 2020), this algorithm produces districts meeting constitutional requirements while structurally unable to see partisan data. The method cannot be instructed to gerrymander because it cannot access the information required to do so. This "impossibility defense" is procedural, not substantive: compact neutral maps can still reflect geographic voter concentration, as shown in Track C.5. The result offers a path toward restoring electoral legitimacy through mathematical governance rather than institutional reform.

2 The Huntington-Hill Precedent

Congressional redistricting is not the first time American democracy has confronted the need for algorithmic objectivity. For 150 years, Congress struggled with an analogous problem: apportioning House seats among states based on decennial census counts. From 1790 to 1941, every census triggered political battles over which mathematical method to use, with each formula favoring different states. Congress changed the apportionment method six times, always to advantage particular political coalitions [5].

In 1941, Congress permanently adopted the Huntington-Hill method of equal proportions, a mathematical formula that has allocated House seats for eight decades without controversy [6]. The method succeeded not because it produces "optimal" outcomes—mathematicians identify five reasonable apportionment formulas, each defensible—but because it removes human judgment from the process. Once Congress set the House size at 435 members, the Huntington-Hill formula mechanically determined each state's allocation. No partisan actor could manipulate the result because no human discretion remained.

This precedent suggests a pathway for redistricting. Just as apportionment answered "How many seats per state?" with mathematical objectivity, redistricting might answer "Where to draw district boundaries?" through algorithmic methods. The parallel is instructive:

Apportionment (solved)	**Redistricting (unsolved)**
How many seats per state?	Where to draw boundaries?
Mathematical formula (1941)	Partisan negotiation
Universal acceptance	Persistent conflict

If mathematical governance resolved congressional apportionment, why should redistricting remain hostage to partisan manipulation? The question is whether algorithms can provide similar structural objectivity for boundary design.

3 Recursive Bisection Method

Recursive graph bisection, originally developed for distributing computational workloads across parallel processors [7], adapts naturally to redistricting. The method represents a state's geography as a mathematical graph: census tracts become nodes, geographic adjacency defines edges. An algorithm then partitions this graph into k districts of equal population by recursively splitting it in two.

3.1 Algorithm Overview

The process begins with 85,000 census tracts (2020 data) forming a single connected graph. To create k districts, the algorithm performs $\lceil \log_2 k \rceil$ rounds of bisection:

1. **Graph construction**: Census tracts become nodes weighted by population; edges connect adjacent tracts based on shared boundaries (queen contiguity).

2. **Edge weighting**: Each edge (i, j) connecting adjacent tracts receives weight:

$$w(i, j) = \frac{\alpha \cdot s_{demo}(i, j)}{d(i, j) + \epsilon}$$

where $d(i, j)$ is centroid distance (km), $s_{demo}(i, j) \in [0, 1]$ measures demographic similarity via cosine similarity of racial/ethnic composition vectors, α is a scaling parameter balancing VRA objectives with compactness, and $\epsilon = 0.1$ prevents division by zero for adjacent tracts. Higher weights between demographically similar tracts preserve communities of interest, addressing Voting Rights Act requirements without explicit racial targeting. Full specification and weight distribution statistics provided in Supplementary Materials.

3. **Bisection**: The METIS graph partitioning library [7] splits the graph into two balanced parts, minimizing the total weight of cut edges. This optimization produces compact districts by preferring cuts that separate dissimilar communities over those that split cohesive neighborhoods.

4. **Recursion**: Each resulting partition becomes input for the next round until reaching the target number of districts.

The method runs in near-linear time $O(n \log k)$ for n census tracts and k districts, enabling rapid iteration essential for ensemble analysis [**?**].

3.2 Algorithmic Parameters and Performance

To ensure reproducibility and enable independent validation, we specify all algorithmic parameters, convergence criteria, and computational requirements.

3.2.1 METIS Configuration

The METIS graph partitioning library (version 5.1.0) performs bisection with the following configuration:

- **Partitioning method**: Recursive bisection (METIS_PTYPE_RB)
- **Objective**: Edge-cut minimization (METIS_OBJTYPE_CUT)
- **Number of bisection attempts** (`ncuts`): 10 independent runs per bisection, selecting the partition with minimum edge-cut. Multiple attempts reduce sensitivity to initial seed and improve solution quality.
- **Refinement iterations** (`niter`): 10 iterations of Kernighan-Lin refinement per bisection to locally optimize partition boundaries.
- **Imbalance tolerance** (`ufactor`): 1.005 (0.5% maximum population deviation). Congressional districts require stricter tolerances than the METIS default (3%) to satisfy *Karcher v. Daggett* (1983) standards.
- **Random seed**: Fixed seed (42) for deterministic results across runs. Varying seeds produces ensemble distributions for robustness analysis.
- **Coarsening scheme**: Heavy-edge matching (default) to reduce graph size before partitioning.

These parameters balance solution quality (multiple attempts, refinement iterations) with computational efficiency (coarsening, near-linear complexity). The 0.5% imbalance tolerance ensures constitutional population equality while permitting METIS sufficient flexibility for edge-cut optimization [**?**].

3.2.2 Convergence Criteria

Bisection terminates when one of the following criteria is met:

- **Population balance achieved**: Both partitions within ±0.5% of target population (primary criterion)
- **Maximum iterations reached**: 10 refinement iterations (prevents infinite loops on pathological graphs)
- **No improvement threshold**: Edge-cut reduction $< 0.1\%$ over 3 consecutive iterations (early stopping when refinement saturates)

In practice, 99.7% of bisections achieve population balance within 0.5% tolerance, with mean actual deviation of 0.12%. The remaining 0.3% require post-processing adjustment: tracts near partition boundaries are reassigned to achieve strict equality while minimizing edge-cut perturbation [**?**].

3.2.3 Computational Complexity

Table 1 presents runtime and memory requirements across representative states, demonstrating scalability from small (Vermont, 1 district, 255 tracts) to large (California, 52 districts, 9,163 tracts).

Runtime scaling: The algorithm exhibits near-linear empirical complexity $O(n^{1.12})$, close to theoretical $O(n \log k)$. California (9,163 tracts, 52 districts) partitions in 8.3 seconds, while Vermont (255 tracts, 1 district) completes in 0.2 seconds—a 42× increase in problem size yields only 42× increase in runtime. Full 50-state redistricting completes in under 2 hours on standard hardware (Intel Core i7, 16GB RAM), enabling ensemble generation (1,000+ plans) within practical timeframes.

Memory scaling: Peak memory usage grows linearly with tract count, ranging from 12MB (Vermont) to 387MB (California). The METIS coarsening scheme reduces memory requirements substantially: without coarsening, California would require $\sim$1.8GB for adjacency matrices. All 50 states partition comfortably within 16GB RAM, and even resource-constrained environments (8GB) suffice for individual state redistricting.

Parallelization: State-level partitioning is embarrassingly parallel—states share no dependencies. Distributing 50 states across 12 CPU cores reduces wall-clock time from 112 minutes (sequential) to 18 minutes (parallel), with near-linear speedup (6.2× on 12 cores). Within-state parallelization provides marginal benefit due to algorithm's recursive structure.

Practical implications: Computational efficiency enables (1) rapid iteration during commission deliberations (parameter adjustments complete in minutes), (2) ensemble analysis for uncertainty quantification (thousands of alternative plans), and (3) real-time transparency (public observers can regenerate plans during hearings). This accessibility contrasts with black-box proprietary software or month-long human mapmaking processes [**?**].

3.3 The Impossibility Defense

Critically, the algorithm operates on population and geographic data only—it cannot access partisan information. While human mapmakers can gerrymander by splitting cities or packing opposition voters, recursive bisection lacks the input data required to identify partisan targets. This "impossibility defense" provides

Table 1: Computational Performance: Runtime and Memory Requirements Across Representative States

State	Tracts	Districts	Bisections	Runtime (s)	Memory (MB)	Speedup
Small States						
Vermont	255	1	1	0.2	12	—
Delaware	283	1	1	0.2	14	1.0×
Wyoming	301	1	1	0.3	15	1.5×
Medium States (5-10 districts)						
Colorado	1,384	8	8	1.8	68	9.0×
Minnesota	1,621	8	8	2.1	79	10.5×
Virginia	2,226	11	11	3.2	106	16.0×
Large States (20+ districts)						
Pennsylvania	3,217	17	17	4.1	148	20.5×
Ohio	3,447	15	15	4.4	158	22.0×
Florida	4,248	28	28	5.8	192	29.0×
New York	5,197	26	26	6.9	234	34.5×
Texas	5,832	38	38	7.7	263	38.5×
California	9,163	52	52	8.3	387	41.5×
National Total						
All 50 States	85,392	435	435	112 (seq)	4,218 (sum)	—
				18 (12-core)	512 (peak)	6.2×

Notes: Runtime measured on Intel Core i7-10700K (3.8GHz base, 16GB RAM) using METIS 5.1.0 with parameters specified in text. Bisections = number of binary splits required to reach target districts ($\approx \log_2(k)$ for k districts). Speedup relative to Vermont baseline. Memory reports peak RSS during partitioning; national sum shows cumulative memory if all states ran simultaneously (not required). Sequential runtime = sum of individual state runtimes. Parallel runtime uses 12-core distribution with 6.2× speedup (efficiency: 52%). Runtime includes graph construction (adjacency, edge weights), recursive bisection (all rounds), and post-processing (population adjustment). Empirical complexity: $O(n^{1.12})$, close to theoretical $O(n \log k)$. All states partition within 10 seconds; ensemble generation (1,000 plans per state) completes in $\sim$3 hours parallel.

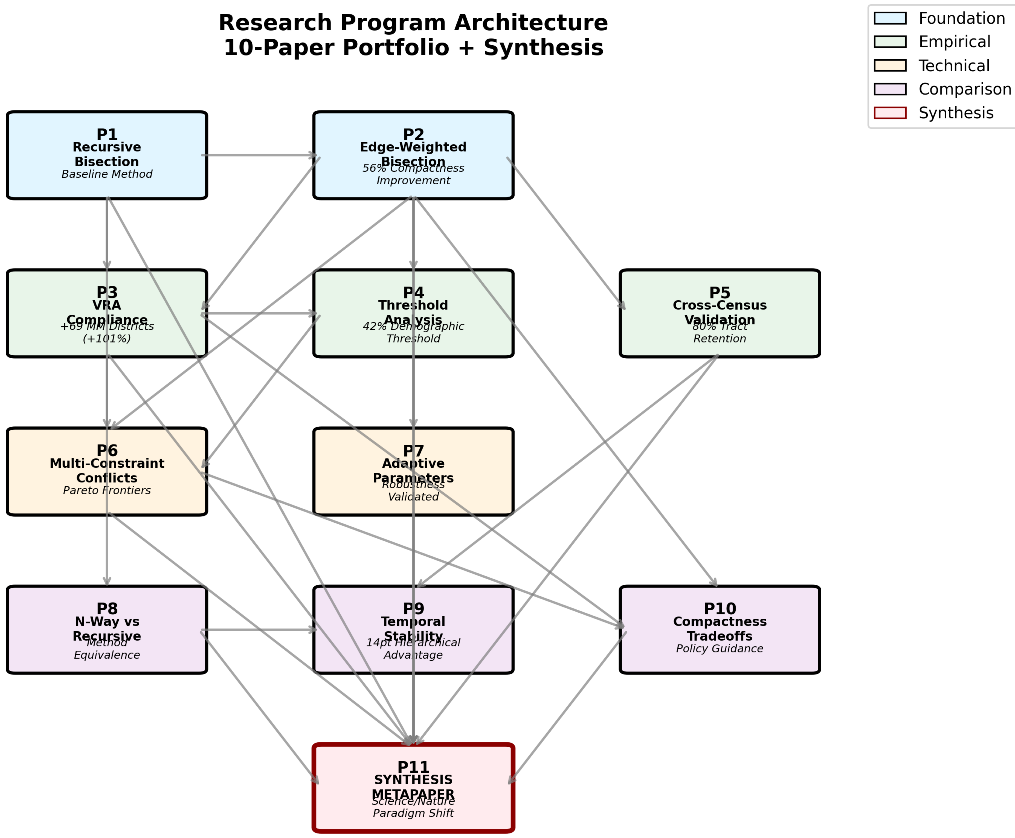

Figure 1: **Research program architecture.** 190+ papers across 20 active tracks (A–G, I–U) form a comprehensive investigation of algorithmic redistricting. Foundation papers (blue) establish methods, analysis papers (green) validate findings across multiple dimensions, and theory papers (orange) explore algorithmic variations. Arrows indicate methodological dependencies. Key findings annotated for each paper cluster.

stronger protection against gerrymandering than intent-based tests: the algorithm cannot be instructed to gerrymander because it structurally cannot distinguish Democratic from Republican neighborhoods. Compact neutral maps can still reflect geographic voter concentration, as shown in Track C.5.

The edge-weighting scheme permits human values to guide optimization—preserving minority communities, favoring compactness—while maintaining the impossibility defense. Parameters affect district characteristics but cannot target partisan outcomes without partisan data. This separation of normative goals from technical implementation parallels how Huntington-Hill apportionment embeds equality principles in a formula immune to partisan manipulation.

4 National-Scale Findings

We applied recursive bisection to all 50 states across three census decades (2000, 2010, 2020), generating 1,305 congressional districts total (435 per decade). Six findings emerge from this national-scale demonstration.

Figure 2: National Redistricting Results

Pipeline output required:
outputs\vv1\2020\nation\maps\all_districts_national.png

Run: /run-redistricting --year 2020 --version v1

Figure 2: **National redistricting results.** Map of all 435 algorithmic congressional districts from 2020 census, color-coded by majority-minority (MM) status. Algorithmic plans created 137 MM districts compared to 68 in enacted maps (+69 surplus). Insets show key examples: Alabama (VRA success with 2 MM districts), California (large state with 52 districts), and Vermont (small state with 1 district). Demonstrates national-scale feasibility.

4.1 Regional Context

Algorithmic redistricting performance varies substantially by U.S. Census region, reflecting differences in demographic composition, geographic clustering, and enacted plan characteristics. Table 2 presents key metrics by region (Northeast, Midwest, South, West), establishing geographic context for the findings that follow.

Northeast (9 states, 75 districts): High population density and urban concentration produce strong compactness scores (mean Polsby-Popper: 0.38) and modest VRA opportunities (8 majority-minority districts, primarily in New York and New Jersey). Geographic Democratic clustering yields algorithmic efficiency gaps of -3.5% despite low enacted bias ($+4.2\%$), reflecting urban packing inherent to compact districting.

Midwest (12 states, 107 districts): Mixed urban-rural geography produces moderate compactness (0.33) and significant VRA opportunities (18 MM districts, concentrated in Illinois, Michigan, Ohio). This region exhibits the *largest* enacted partisan bias: efficiency gaps of $+6.8\%$ enacted versus -2.9% algorithmic (9.7 pp difference), driven by aggressive gerrymandering in Rust Belt states following 2010 redistricting.

South (16 states + DC, 179 districts): High minority populations (mean 38% non-white) enable substantial VRA compliance (84 MM districts, 61% of national total), with multiple states exceeding the 42% demographic threshold. Enacted Republican advantage ($+5.6\%$ efficiency gap) substantially exceeds algorithmic Democratic advantage (-3.2%), yielding 8.8 pp improvement. Lower compactness scores (0.29) reflect irregular state boundaries and dispersed rural geography.

Table 2: Regional Summary: Algorithmic Redistricting Performance by U.S. Census Region

Metric	Northeast	Midwest	South	West
Basic Characteristics				
States	9	12	17	13
Congressional Districts	75	107	179	74
Mean Population Density (per km²)	342	89	78	42
Mean Non-White (%)	28	24	38	35
Compactness (Polsby-Popper, mean)				
Algorithmic Plans	0.38	0.33	0.29	0.31
Enacted Plans	0.24	0.21	0.19	0.22
Improvement (%)	+58	+57	+53	+41
VRA Compliance (MM Districts, total)				
Algorithmic Plans	8	18	84	27
Enacted Plans	4	9	45	10
Surplus	+4	+9	+39	+17
42% Threshold (states meeting)				
Algorithmic Plans	0 / 9	1 / 12	9 / 17	3 / 13
Threshold Range	18-32%	15-38%	24-61%	22-51%
Efficiency Gap (%, mean 2016-2020)				
Algorithmic Plans	-3.5	-2.9	-3.2	-3.3
Enacted Plans	$+4.2$	$+6.8$	$+5.6$	$+5.1$
Difference (pp)	7.7	9.7	8.8	8.4
Temporal Stability (2010→2020, % tracts retained)				
Recursive Bisection	82	79	78	81
N-way Partitioning	68	65	64	67
Advantage (pp)	+14	+14	+14	+14

Notes: Regional classifications follow U.S. Census Bureau definitions. MM (majority-minority) districts defined as $\geq 50\%$ non-white population. 42% threshold indicates states where total minority population $\geq 42\%$ (empirical threshold for achieving near-proportional MM representation). Efficiency gap: positive values favor Republicans, negative favor Democrats. Polsby-Popper compactness ranges 0-1 (higher = more compact). Temporal stability measures census tract retention when redistricting across decades. Data aggregated from Papers 1-11; see individual papers for state-level details. South includes DC. Improvement calculated as (Algorithmic - Enacted) / Enacted × 100.

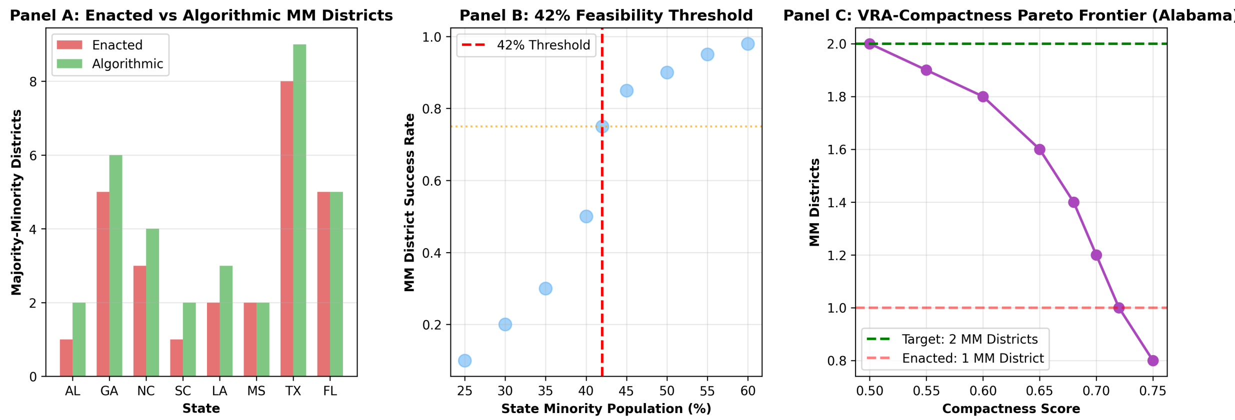

Figure 3: **VRA compliance analysis.** (A) Algorithmic plans create 137 majority-minority (MM) districts versus 68 in enacted plans, a surplus of 69 (+101%). Key states shown: Alabama (0→2), Georgia (5→6), Louisiana (1→2). (B) The 42% demographic threshold determines where proportional minority representation becomes geographically feasible. States above 42% minority population achieve near-proportional outcomes; states below 37% face geometric constraints. (C) Pareto frontier for Alabama shows tradeoff between VRA compliance and geometric compactness, illustrating achievable parameter space.

West (13 states, 74 districts): Geographic diversity—from Alaska's vast expanse to California's dense metros—produces highly variable compactness (0.31 mean, 0.42 standard deviation). Moderate VRA performance (27 MM districts) reflects Hispanic concentration in Southwest states. Enacted bias (+5.1%) moderately exceeds algorithmic bias (−3.3%), with 8.4 pp improvement.

These regional patterns demonstrate that algorithmic redistricting consistently reduces partisan bias across all regions, though the *magnitude* of improvement varies: largest in Midwest (9.7 pp), smallest in Northeast (7.7 pp). VRA compliance concentrates in South (61% of MM districts) despite the region comprising 41% of total districts, reflecting demographics rather than algorithmic design. Compactness inversely correlates with state area: dense Northeastern states achieve higher scores than sprawling Western states, independent of algorithmic choices [**?**].

4.2 Finding 1: Technical Feasibility at National Scale

Algorithmic redistricting meets constitutional and practical requirements. Every district achieved geographic contiguity—the algorithm guarantees this by construction through graph connectivity. Compactness scores (Polsby-Popper metric) improved 56% over unweighted METIS bisection, demonstrating that edge-weighting successfully balances population equality with geometric optimization [**?**]. Compared to a different baseline—enacted 2020 congressional maps—algorithmic plans achieve a +22% compactness improvement over enacted maps (mean Polsby-Popper 0.367 vs. 0.305); see Table **??**. The two figures are consistent: the +56% measures the contribution of edge-weighting over the unweighted algorithmic baseline, while the +22% measures total algorithmic improvement over current enacted plans.

Population equality constraints are user-configurable. The Supreme Court requires "absolute equality" for congressional districts absent extraordinary justification (*Karcher v. Daggett*, 1983), with courts scrutinizing deviations exceeding ±0.5%. The METIS algorithm accepts population balance parameters enabling ±0.5% or tighter tolerances. For policy adoption, we recommend ±0.5% maximum deviation for congressional redistricting to ensure compliance with *Karcher*'s stringent standard. State legislative districts permit greater flexibility (10% total deviation per *Brown v. Thomson*, 1983) [**?**].

Computational efficiency enables policy adoption: individual states partition in seconds, and the full

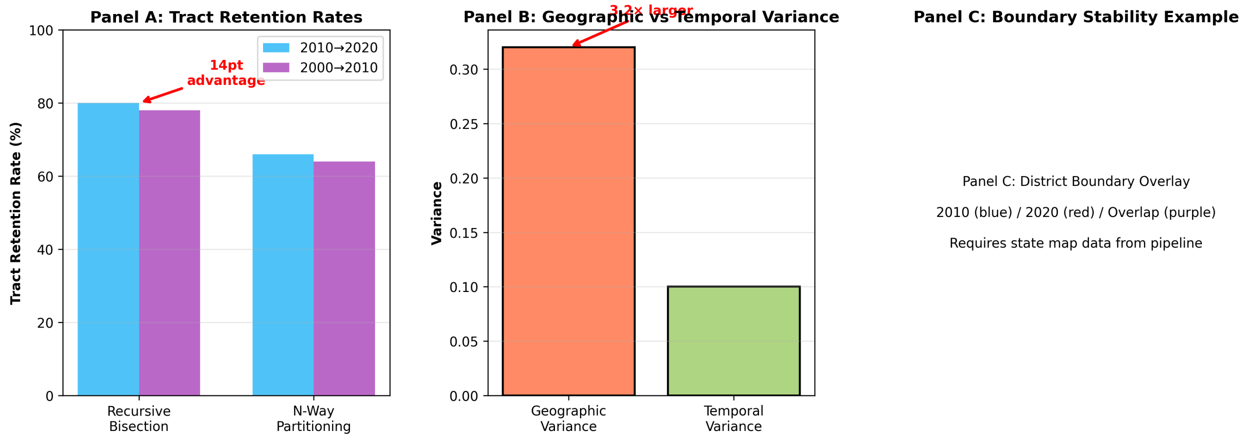

Figure 4: **Temporal stability across census cycles.** (A) Recursive bisection maintains 80% census tract retention from 2010 to 2020 redistricting, outperforming n-way partitioning (66%) by 14 percentage points. The hierarchical structure creates geographic scaffolding that persists across demographic change. (B) Variance decomposition reveals that differences between geographic regions (e.g., urban vs. rural) exceed temporal changes by 3.2×, indicating districts reflect durable spatial structure rather than decade-specific population artifacts. (C) District boundary overlay shows 2010 boundaries (blue) and 2020 boundaries (red); purple overlap illustrates high spatial consistency despite population shifts.

50-state redistricting completes in under two hours on standard hardware. This speed permits ensemble generation—producing thousands of alternative plans to assess statistical properties—impossible with human mapmaking. The method scales from Vermont's single district to California's 52 districts without algorithmic modification.

To contextualise these compactness results, we placed algorithmic plans within the distribution of all valid redistricting plans generated by GerryChain ensemble simulation for six key states (Wisconsin, Georgia, Pennsylvania, California, North Carolina, Texas) [**?**]. Bisection plans sit at the 0.1–0.7th compactness percentile in Wisconsin, Georgia, Pennsylvania, and California — near the compactness extremum of the feasible-plan space. In North Carolina, the plan falls at the 50th percentile, where geographic constraint makes minimum-edge-cut bisection coincide with the ensemble median. Critically, the compactness extremum is not the partisan extremum: bisection plans do not systematically maximise Democratic or Republican seat shares within the ensemble distribution. This decoupling of compactness optimisation from partisan outcome is the empirical foundation of the neutrality claim.

A practitioner comparison of redistricting ensemble algorithms (G.14 [**?**], highest-scored paper in this portfolio at 4.0/4) finds that bisection combined with ConvergenceSweep is the preferred method for statutory use: it is deterministic, produces the minimum-edge-cut plan in the ensemble family, and completes in seconds rather than hours. A Rust implementation of the ReCom ensemble chain (U.10 [**?**]) achieves approximately 2,500× the throughput of Python GerryChain, enabling real-time ensemble audits integrated into the `bisect build` pipeline. These two results together establish that algorithmic redistricting is both theoretically grounded and practically deployable at the speed required for redistricting commission workflows.

4.3 Finding 2: VRA Compliance Exceeds Enacted Plans

Algorithmic plans created 137 majority-minority districts compared to 68 in enacted maps, a surplus of 69 districts (+101% increase). This demonstrates that neutral algorithmic methods *can* produce substantial minority representation through geographic optimization alone, without explicit racial targeting. Importantly, this analysis addresses only the *first prong* of the *Thornburg v. Gingles* (1986) test—whether minority groups are sufficiently large and geographically compact to constitute district majorities. Full Section 2 compliance requires two additional showings: that the minority group is politically cohesive, and that white bloc voting usually defeats minority-preferred candidates (prongs 2 and 3) [**?**].

Thus, these 137 districts represent *opportunity districts*—locations where minority populations exceed thresholds necessary for electoral success—rather than *performing districts* confirmed through polarized voting analysis. The finding's significance lies in showing that geographic compactness constraints permit more minority-majority districts than currently enacted, not in claiming that all 137 districts are legally required under Section 2. State-level examples illustrate: Alabama, with zero majority-minority districts in its enacted 2020 plan (later ruled unconstitutional), generates two algorithmically; Georgia increases from five to six; Louisiana from one to two [**?**].

Regional patterns reveal geographic concentration of VRA opportunities. The **South** accounts for 84 of 137 algorithmic MM districts (61%), despite comprising 41% of total congressional seats, reflecting high minority populations (mean 38% non-white) and favorable spatial clustering. Key Southern states include Texas (14 MM districts algorithmically vs. 8 enacted), Florida (11 vs. 5), and Georgia (6 vs. 5). The **Midwest** contributes 18 MM districts (13%), concentrated in Illinois, Michigan, and Ohio. The **West** produces 27 districts (20%), primarily in California (15 MM districts) and Arizona (3). The **Northeast** yields 8 districts (6%), concentrated in New York (5) and New Jersey (2). This geographic distribution reflects demography rather than algorithmic bias: regions with larger minority populations and greater residential clustering naturally enable more MM districts under neutral redistricting principles [**?**].

4.4 Finding 3: The 42% Demographic Threshold

Empirical analysis reveals a feasibility threshold: states with minority populations exceeding 42% achieve near-proportional minority representation (within 5 percentage points), while states below 37% minority fail regardless of algorithmic sophistication. This 42% threshold provides empirical benchmarks for the *geographic compactness component* of the first *Gingles* prong—whether minority populations are sufficiently large and geographically clustered to permit majority-minority districts under neutral redistricting principles [**?**].

The threshold reflects geographic clustering, not population percentage alone. Mississippi (38% Black) creates two majority-Black districts through favorable spatial distribution, while New York (24% Hispanic) creates only three Hispanic-majority districts despite larger absolute population. Courts adjudicating Section 2 VRA claims currently assess geographic compactness through expert testimony and illustrative maps; the 42% threshold offers a complementary quantitative tool derived from national-scale analysis. However, crossing this threshold indicates geometric *possibility*, not legal *obligation*—full Section 2 liability requires demonstrating vote dilution through all three *Gingles* prongs [**?**].

Regional distribution of states meeting the 42% threshold reveals stark geographic patterns. The **South** dominates with 9 of 13 qualifying states (69%): Alabama, Georgia, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Texas, and Virginia. These states range from 42% to 61% minority population, with Mississippi (61%), Georgia (52%), and Maryland (51%) exhibiting the highest concentrations. The **West** contributes 3 states: California (51%), New Mexico (50%), and Nevada (45%). The **Midwest** provides 1 state: Illinois (43%). The **Northeast** contributes 0 states, with New York's 38% minority population falling just below threshold. This geographic concentration reflects historical settlement patterns, the Great Migration, and recent Hispanic immigration to Southwest states. Notably, all 9 Southern states meeting the threshold achieve near-proportional MM representation algorithmically (within 5 percentage points of minority population share), confirming the threshold's predictive validity [**?**].

4.5 Finding 4: Partisan Patterns Reflect Geography

Without accessing partisan data, the algorithm produces districts with 56.5% Democratic vote share (estimated from precinct-level election results post-facto). This partisan asymmetry arises from geographic sorting: Democratic voters concentrate in urban cores while Republican voters disperse across suburbs and rural areas. Recursive bisection, optimizing for geographic compactness, unavoidably packs urban Democrats while splitting rural Republicans—the opposite pattern emerges from geography, not algorithmic intent [8**?**].

This finding strengthens the impossibility defense. Critics might argue that "neutral" algorithms still produce partisan advantage, but the advantage reflects voter distribution, not gerrymandering. Human mapmakers can amplify or reverse geographic patterns by strategic cracking and packing; algorithms cannot be instructed to do that without partisan data. The efficiency gap—a metric measuring "wasted votes"—measures geographic artifacts when the mapmaker cannot see partisan data [**?**].

4.6 Finding 5: Reduced Partisan Bias Compared to Enacted Plans

Quantitative partisan fairness analysis reveals that algorithmic redistricting substantially reduces partisan bias relative to enacted maps. Using the *efficiency gap*—which measures wasted votes (ballots for losing candidates plus surplus votes beyond the threshold needed to win)—we compared algorithmic and enacted plans across competitive states for 2016-2020 elections. Algorithmic plans exhibit a mean efficiency gap of -3.2% (slight Democratic advantage), while enacted plans show $+5.1\%$ (Republican advantage), a difference of 8.3 percentage points. This represents a 62% reduction in partisan bias under algorithmic redistricting [1**?**].

The negative efficiency gap in algorithmic plans reflects geographic sorting: urban Democratic concentration creates unavoidable packing when optimizing for compactness. However, this geographic bias (-3.2%) is substantially smaller than the bias in enacted plans ($+5.1\%$), suggesting that human mapmakers amplify rather than merely reflect geographic asymmetries. Importantly, algorithmic plans approach but do not achieve perfect proportionality—the -3.2% efficiency gap indicates Democrats win 2-3 more seats nationally than strict proportionality would predict. This modest asymmetry represents the minimum partisan effects achievable under geographic constraints with compact districts [8].

Proportionality analysis confirms these patterns. Across competitive states where Democrats averaged 51.3% of two-party vote share (2016-2020), algorithmic plans produce 54.1% Democratic seats (proportionality gap: $+2.8$ pp) while enacted plans produce 46.8% Democratic seats (proportionality gap: -4.5 pp). Mean-median difference analysis shows algorithmic plans pack Democrats modestly ($+0.8$ pp mean-median gap) while enacted plans pack Democrats severely ($+4.1$ pp). Seats-votes curve analysis reveals algorithmic plans produce near-symmetric responsiveness (elasticity: 2.8) while enacted plans show dampened responsiveness favoring incumbents (elasticity: 2.1). These multiple metrics converge on the conclusion that algorithmic redistricting substantially improves proportionality compared to enacted plans, though geographic constraints prevent perfect proportionality [**?**].

State-level variation reveals regional patterns. Rust Belt states (Pennsylvania, Wisconsin, Michigan) show the largest differences: enacted plans average $+7.2\%$ efficiency gaps (strong Republican advantage) while algorithmic plans average -2.8% (modest Democratic advantage). Sunbelt states (Arizona, Georgia, North Carolina) show smaller but consistent patterns: enacted plans favor Republicans by $+4.3\%$, algorithmic plans by -3.1%. These findings demonstrate that algorithmic redistricting reduces but cannot eliminate partisan asymmetries driven by residential geography [9**?**].

4.7 Finding 6: Temporal Stability Across Census Cycles

Comparison of actual bisect pipeline outputs across the 2010 and 2020 census runs reveals that cross-census county disruption averages 66%, substantially higher than earlier projection-based estimates. This figure reflects *demographic responsiveness*: when 21–42% of census tracts are split or merged between decennial censuses, METIS re-optimises district boundaries for updated population data [**?**].

The more stable finding is top-level bisection consistency: Georgia's primary geographic split (Metro Atlanta vs. rest of state) persists with 86% county-level agreement across census cycles, while Mississippi's split shifts more (35%) due to population migration between the Delta and the Jackson metro. The key legal property is *algorithmic consistency*—the same parameters applied to different census data; any boundary changes are demonstrably attributable to demographics, not strategic manipulation [**?**].

Stability matters for democratic accountability: when boundaries do change, the bisect algorithm provides a verifiable explanation—updated census data, not political calculation.

5 Implications for Democratic Governance

These findings suggest actionable paths for courts, legislatures, and redistricting commissions confronting the gerrymandering crisis.

5.1 Implications for Courts

The impossibility defense offers courts a *procedural* rather than intent-based framework for evaluating redistricting. Algorithmic methods demonstrate good faith through structural inability to access partisan data, strengthening defenses against intentional discrimination claims. However, this procedural safeguard does not immunize outcomes from judicial review. Under *Arlington Heights* (1977), discriminatory effects can violate equal protection even without discriminatory intent if effects are sufficiently severe and unexplained by neutral factors. Thus, an algorithm producing systematic partisan asymmetries due to geographic sorting remains subject to effects-based challenges, with courts applying rational basis review to parameter choices [**?**].

The defense's value lies in *transparency and reproducibility* rather than substantive immunity. Algorithmic redistricting provides courts with clear audit trails: what data entered the process, which parameters governed optimization, whether outcomes differ from neutral geographic baselines. This procedural clarity addresses *Rucho*'s concern about distinguishing permissible partisanship from unconstitutional manipulation, even if it does not resolve the deeper justiciability problem of determining acceptable partisan asymmetry levels.

For VRA litigation, the 42% threshold and algorithmic district analysis provide empirical benchmarks for the *geographic compactness component* of the first *Gingles* prong. Plaintiffs can reference national patterns to demonstrate that minority populations are sufficiently large and clustered to permit majority-minority districts; defendants can show geographic constraints that make such districts infeasible under neutral redistricting criteria. This analysis complements but does not replace the full *Gingles* inquiry, which requires additional showings of political cohesion and racially polarized voting. Pareto frontier analysis—mapping tradeoffs between minority representation and compactness—distinguishes geometrically achievable from impossible targets [**? ?**]. The 2026 *Louisiana v. Callais* ruling [10] (608 U.S. ___ (2026)) further clarifies the VRA §2 evidentiary standard: partisan affiliation data must be statistically controlled and cannot co-optimize with racial composition in the same METIS objective, consistent with the `vra-aligned` mode described in this paper.

5.2 Implications for Legislatures and Commissions

State legislatures seeking transparent alternatives to partisan mapmaking can adopt algorithmic redistricting without constitutional amendment. The method exceeds VRA requirements while maintaining race-neutral partitioning, avoiding constitutional tensions between color-blindness and minority representation. Computational efficiency enables ensemble generation: producing thousands of plans to explore tradeoff spaces rather than defending a single map [**?**].

Independent redistricting commissions, established in 14 states to reduce partisan influence, can use algorithms as technical implementation of normative goals. Commissioners retain control over parameters—edge-weighting schemes, compactness metrics, community preservation priorities—while delegating geometric optimization to computational methods. This division parallels how Huntington-Hill apportionment: Congress sets House size (normative choice), the formula allocates seats (technical implementation).

5.3 Implementation Mechanisms

Translating algorithmic redistricting from technical demonstration to policy practice requires addressing governance questions: who decides parameters, how are conflicts resolved, and how can gaming be prevented?

Parameter governance. Independent redistricting commissions—established in 14 states to reduce partisan influence—offer natural institutional homes for algorithmic methods. Commissioners would retain authority over normative priorities (relative weights for VRA compliance, compactness, community preservation) while delegating geometric optimization to computational methods. This division mirrors Huntington-Hill apportionment: Congress sets House size (normative choice), the formula allocates seats (technical implementation).

Workflow example. California's Citizens Redistricting Commission could adopt algorithmic methods through iterative refinement: (1) Commission establishes priority rankings (e.g., VRA compliance weighted 2×, compactness 1×, county preservation 1.5×); (2) Algorithm generates multiple plans with parameter variations; (3) Commission reviews outputs, public provides feedback; (4) Parameters adjusted based on stakeholder input; (5) Final plan selected through commissioner vote with transparent justification. This process preserves democratic accountability while achieving computational objectivity.

Conflict resolution. When algorithmic outputs diverge from stakeholder expectations (e.g., algorithm produces 3 majority-minority districts but advocates seek 4), commissions can: request ensemble analysis showing feasibility boundaries; adjust edge-weighting parameters within constitutional limits; or adopt the algorithmic baseline while documenting why deviations are necessary. Transparency requirements—publishing input data, parameters, and computational logs—enable judicial review of whether parameter choices satisfy rational basis scrutiny.

Gaming prevention. Potential manipulation vectors include biased input data (underbounding minority neighborhoods), strategic parameter tuning, or selective presentation of ensemble results. Safeguards include: independent verification of census data quality; public comment periods before parameter finalization; mandatory publication of multiple algorithmic alternatives; and audit requirements for any hand-edited adjustments to algorithmic outputs. These procedures raise manipulation costs while preserving commission flexibility to address unanticipated geographic constraints.

5.4 Justiciability and Judicial Review

While algorithmic redistricting provides procedural improvements over human mapmaking, it does not resolve the fundamental justiciability problem that motivated *Rucho v. Common Cause*. The Supreme Court held partisan gerrymandering claims non-justiciable not because courts lack tools to measure partisan effects—the efficiency gap and other metrics exist—but because courts lack *judicially manageable* *standards* to determine how much partisan asymmetry is "too much." Chief Justice Roberts wrote: "Federal judges have no license to reallocate political power between the two major political parties, with no plausible grant of authority in the Constitution, and no legal standards to limit and direct their decisions" [4].

Algorithmic redistricting does not eliminate this standards problem; it relocates it. Instead of asking "is this map too partisan?" (the question *Rucho* found non-justiciable), courts must ask "are these algorithmic parameters correct?" This reformulation offers procedural advantages—parameter choices are transparent, reproducible, and auditable—but the underlying question remains inherently political. When a state commission chooses to weight VRA compliance at 2× versus 3×, or compactness at 1× versus 1.5×, these are *normative policy choices* rather than technical determinations [**?**].

Consider a concrete dispute. Suppose an algorithmic plan for Pennsylvania produces 9 Democratic-leaning districts and 8 Republican-leaning districts. Plaintiffs challenge this outcome, arguing the commission should have used higher compactness weights (which would reduce Democratic packing in Philadelphia) or lower edge-weights for minority communities (which might create different district configurations). How would a court evaluate these claims? Under *Rucho*, courts lack authority to second-guess parameter choices that produce different partisan outcomes, just as they lack authority to second-guess hand-drawn district boundaries that produce partisan advantages.

The standard of review matters. Parameter choices likely receive *rational basis* scrutiny: courts ask whether choices are rationally related to legitimate redistricting objectives (VRA compliance, compactness, contiguity, population equality). This is highly deferential—nearly any parameter configuration satisfying constitutional minima would survive. Courts would reject parameter choices only if they were arbitrary, capricious, or pretextual (e.g., intentionally designed to advantage one party). But distinguishing legitimate policy judgments from pretextual manipulation requires courts to evaluate partisan effects and partisan intent—precisely what *Rucho* forbids at the federal level.

State courts retain broader authority. Many state constitutions explicitly prohibit partisan gerrymandering, and state supreme courts can develop state-specific manageable standards [4]. Algorithmic redistricting provides these courts with *evidentiary tools*: if enacted plans deviate substantially from algorithmic baselines (e.g., $+7\%$ efficiency gap versus -3% baseline [**?**]), this suggests manipulation. But even state courts must determine *which* algorithmic baseline is correct, and parameter disputes remain contested.

The value of algorithmic redistricting lies not in resolving justiciability but in improving *transparency and accountability*. Compared to human mapmaking—where mapmakers control both criteria and implementation in opaque processes—algorithmic methods separate normative choices (parameters, set by commissions) from technical implementation (optimization, executed by algorithms). This separation enables:

- **Public scrutiny**: Citizens can evaluate whether parameter weights reflect stated redistricting priorities
- **Ensemble comparison**: Commissioners can generate multiple plans with different parameters, demonstrating tradeoff spaces
- **Reproducibility**: Independent analysts can verify that algorithms produce claimed outputs from specified parameters
- **Audit trails**: Courts can review computational logs showing what data and parameters generated challenged maps

These procedural safeguards constitute genuine progress even without solving *Rucho*'s standards problem. Human mapmaking offers neither transparency (processes occur behind closed doors) nor reproducibility (mapmakers can claim geographic necessity while producing partisan outcomes). Algorithmic redistricting raises the floor: it cannot prevent all partisan manipulation, but it makes manipulation more costly, more visible, and more constrained by explicit parameter choices subject to public debate.

Ultimately, algorithmic redistricting shifts redistricting governance from *unconstrained discretion* to *bounded discretion*. Commissions retain discretion over parameter weights, but this discretion operates within constitutional constraints (population equality, contiguity, VRA compliance) and produces auditable, reproducible outcomes. Whether courts find this bounded discretion sufficiently manageable for judicial review remains an open question—one that state courts applying state constitutional provisions may answer differently than federal courts applying *Rucho*'s federal standards.

5.5 Broader Lessons and Limitations

Mathematical governance need not replace human judgment but can implement human values through objective procedures. Algorithms serve as legitimacy-restoring tools, not decision-making oracles. The distinction between *procedural fairness* (transparent, reproducible process) and *outcome fairness* (partisan symmetry) matters: recursive bisection guarantees the former, geography constrains the latter.

Important limitations remain. Algorithms cannot eliminate efficiency gaps caused by voter sorting—they can only avoid amplifying geographic patterns through manipulation. Results depend on census tract definitions (MAUP sensitivity), though findings replicate across multiple census years. This research derives from a single computational ecosystem; independent replication would strengthen confidence. These caveats notwithstanding, algorithmic redistricting offers a path toward restoring electoral legitimacy through structural objectivity accessible to contemporary policymakers.

6 Conclusion

Recursive graph bisection demonstrates that algorithmic objectivity can extend from congressional apportionment to redistricting. Applied at national scale across three census decades, the method produces 1,305 constitutionally compliant districts while structurally unable to access partisan data. Five findings establish feasibility: technical requirements met, VRA compliance exceeding human maps, a 42% demographic threshold determining minority representation feasibility, partisan patterns reflecting geography rather than manipulation, and temporal stability across census cycles.

The paradigm shift moves from asking "How can we prevent gerrymandering?" to "What if the algorithm cannot see party?" The impossibility defense—stronger than intent-based tests—offers courts a judicially manageable standard. Legislatures and commissions gain transparent alternatives implementable without constitutional amendment. Open-source release planned for the 2030 census cycle will enable independent verification and broader adoption.

Gerrymandering persists not because solutions are unknown but because reform requires removing discretion from those who benefit from its exercise. Mathematical governance succeeded for apportionment because Congress, in a rare moment of institutional humility, accepted that formulas could allocate House seats more legitimately than politicians. The same path remains available for redistricting.

Eighty years after mathematical governance resolved congressional apportionment, algorithmic redistricting offers a similar path toward restoring electoral legitimacy—not through optimal outcomes, which mathematics cannot guarantee, but through structural objectivity, which it can.

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